WORKED SOLUTION

STEP I 2000, Question 14

Probability and Statistics • Moments of Uniform Distributions

STEP I 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q14

The key idea

Everything here is a moment of a uniform distribution, and the moments of X on $[-1,1]$ are easy: the odd ones vanish by symmetry and the even ones are $1/(k+1)$. For $Z = Y - X$ we don't need the distribution of Z at all. Expanding $(Y-X)^2$ and $(Y-X)^4$ binomially and using independence turns every expectation into a product of moments we already know.

Solution

Moments of X

X has density $\frac{1}{2}$ on $[-1,1]$, so for a whole number k

$$\mathrm{E}(X^k) = \frac{1}{2}\int_{-1}^{1} x^k \,\mathrm{d}x = \begin{cases} 0, & k \text{ odd (the integrand is odd)}, \\ \dfrac{1}{k+1}, & k \text{ even}. \end{cases}$$

In particular $\mathrm{E}(X) = \mathrm{E}(X^3) = 0$, $\mathrm{E}(X^2) = \frac{1}{3}$ and $\mathrm{E}(X^4) = \frac{1}{5}$. Hence

$$\mathrm{E}(X^2) = \frac{1}{3}, \qquad \mathrm{Var}(X^2) = \mathrm{E}(X^4) - (\mathrm{E}(X^2))^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}.$$

Y has the same distribution, so it has the same moments.

The mean of Z^2

Expanding, $Z^2 = Y^2 - 2XY + X^2$. Since X and Y are independent, $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y) = 0$, so

$$\mathrm{E}(Z^2) = \frac{1}{3} - 0 + \frac{1}{3} = \frac{2}{3}.$$

The variance of Z^2

We need $\mathrm{E}(Z^4)$. By the binomial theorem,

$$Z^4 = (Y - X)^4 = Y^4 - 4Y^3X + 6Y^2X^2 - 4YX^3 + X^4.$$

Ask yourself Can we really take the expectation of Y^3X as $\mathrm{E}(Y^3)\,\mathrm{E}(X)$?

Yes: because X and Y are independent, so are any functions of them, such as Y^3 and X, and the expectation of a product of independent random variables is the product of the expectations. Each of the two middle cross terms contains an odd power of X or of Y, so its expectation is zero. Therefore

$$\begin{aligned}\mathrm{E}(Z^4) &= \mathrm{E}(Y^4) + 6\,\mathrm{E}(Y^2)\,\mathrm{E}(X^2) + \mathrm{E}(X^4)\\ &= \frac{1}{5} + 6\cdot\frac{1}{3}\cdot\frac{1}{3} + \frac{1}{5} = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}.\end{aligned}$$

So

$$\mathrm{Var}(Z^2) = \mathrm{E}(Z^4) - (\mathrm{E}(Z^2))^2 = \frac{16}{15} - \frac{4}{9} = \frac{48-20}{45} = \frac{28}{45}.$$

Since $\frac{28}{45} = 7 \times \frac{4}{45}$, we have $\mathrm{Var}(Z^2) = 7\,\mathrm{Var}(X^2)$, as required.

Checks

- Z has the triangular density $f(z) = \frac{1}{4}(2 - |z|)$ on $[-2, 2]$. Directly, $\mathrm{E}(Z^4) = 2\int_0^2 \frac{1}{4}z^4(2-z)\,\mathrm{d}z = \frac{1}{2}\left(\frac{64}{5} - \frac{32}{3}\right) = \frac{16}{15}$, agreeing with the expansion.
- $\mathrm{E}(Z^2) = \mathrm{Var}(Y - X) = \mathrm{Var}(Y) + \mathrm{Var}(X) = \frac{1}{3} + \frac{1}{3}$, since $\mathrm{E}(Z) = 0$.
- A simulation of a million pairs (X, Y) gave $\mathrm{Var}(X^2) \approx 0.0889$ and $\mathrm{Var}(Z^2) \approx 0.623$, against $4/45 \approx 0.0889$ and $28/45 \approx 0.622$.

Common slips

- Writing $\mathrm{Var}(Z^2) = \mathrm{Var}(Y^2) + \mathrm{Var}(X^2)$. This ignores the term $-2XY$ inside Z^2, which has mean zero but contributes $4\,\mathrm{Var}(XY) = \frac{4}{9} = \frac{20}{45}$; the slip gives $\frac{8}{45}$ instead of $\frac{28}{45}$.
- Forgetting the binomial coefficient 6 on the Y^2X^2 term.
- Using density 1 instead of $\frac{1}{2}$ on $[-1, 1]$, which doubles every moment.

Beyond STEP

The quantities $\mathrm{E}(X^k)$ are the moments of X, and the neat way to handle sums of independent variables is the moment generating function $M_X(t) = \mathrm{E}(\mathrm{e}^{tX})$, for which $M_{Y-X}(t) = M_Y(t)M_X(-t)$. Expanding this product as a power series in t gives every

moment of Z at once. Moment and characteristic functions are central in a first probability course: they are the main tool in the proof of the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, develops them clearly.

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